

Pairs Trading / Statistical Arbitrage Playbook

10:30-14:30 ET advanced execution manual for expert traders

Strategy focus: market-neutral relative-value mean reversion after the opening price-discovery shock

Execution focus: spread construction, hedge-ratio discipline, cost-adjusted entries, and portfolio risk controls

Core thesis

Pairs trading is a disciplined relative-value process: identify two related instruments, model the hedge ratio, measure the residual spread, and trade temporary dislocations only when convergence is statistically credible and executable after all frictions.

The 10:30-14:30 window is designed to avoid the noisiest opening repricing and the closing auction/benchmark-flow regime. It is a quality-control window, not a guarantee.

This playbook defines pair selection, spread construction, signal thresholds, execution rules, risk limits, regime filters, backtest protocol, and live-desk checklists.

Professional PDF playbook

Built for advanced and expert traders using minute/tick data, midquotes, borrow data, factor models, and paired execution.

Educational material only. Validate independently before risking capital.

Table of Contents

Section	What it covers
1. Strategy Mandate	Objective, market posture, holding period, and invalidation.
2. Why the 10:30-14:30 Window Exists	Time-of-day rationale and desk clock.
3. Research Basis and Professional Context	Academic evidence and public professional-practice context.
4. The Strategy in One Page	Compact architecture, required stack, and decision matrix.
5-8. Universe, Pair Selection, Spread, Z-Score	Universe design, pair selection, hedge ratio, and signal thresholds.
9-12. Entry, Execution, Management, Stops	Entry patterns, paired execution, exits, and invalidation logic.
13-15. Portfolio, Factor/Event, Variants	Risk-neutrality, regime filters, and strategy variants.
16-19. Backtesting, Data, Schedule, Example	Validation, data engineering, live workflow, and practical example.
20-24. Failure Modes, Enhancements, Checklists, Parameters, References	Operational safeguards, improvement path, checklists, defaults, and sources.

Scope

This playbook is written for U.S. equities, ETFs, and liquid equity-index futures in the 10:30 a.m. to 2:30 p.m. ET window. The framework can be adapted to other markets only after instrument-specific testing.

Risk notice

Pairs trading can appear low-risk because it is market-neutral, but losses can be directional, sudden, and correlated when the relationship breaks. Transaction costs, latency, borrow, corporate actions, factor shocks, and execution legging can destroy edge.

1. Strategy Mandate

Pairs trading / statistical arbitrage is a relative-value mean-reversion strategy. The trader buys the instrument that is cheap relative to a modeled relationship and sells the instrument that is rich. The objective is not to predict the market. The objective is to capture convergence of a residual spread while keeping unwanted beta, sector, and factor exposure controlled.

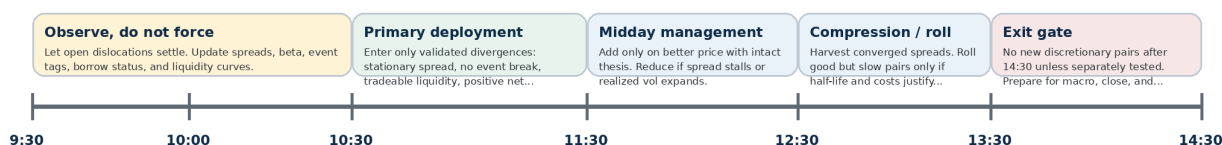
Core rule

A spread is tradable only when the relationship is economically logical, statistically stable, clean from event contamination, and executable after all costs.

Mandate component	Professional standard
Primary execution window	10:30-14:30 ET. Use the open for observation and the final 90 minutes for harvesting, risk reduction, or only separately tested signals.
Market posture	Market-neutral or factor-neutral. Directional exposure should be incidental, measured, and capped.
Primary instruments	Liquid same-sector stocks, ETF pairs, stock-vs-sector ETF residuals, index ETF/futures relationships, and basket residuals.
Holding period	Intraday to multi-day depending on half-life. This PDF focuses on intraday deployment with optional overnight roll rules.
Edge source	Temporary divergence between economically linked instruments, often supported by liquidity imbalance, investor inattention, index/factor flows, or non-fundamental dislocation.
Invalidation	Earnings/news repricing, unstable hedge ratio, broken stationarity, borrow shock, liquidity breakdown, or correlated portfolio drawdown.

Pairs Trading / Statistical Arbitrage - 10:30 to 14:30 ET Operating Clock

Purpose: deploy relative-value mean reversion after opening price discovery has stabilized, before closing benchmark and auction flows dominate.



Desk principle Pairs/stat-arb is not a chart trade. It is a spread-quality trade. The execution window is useful only when the spread is stationary, the catalyst set is clean, hedge ratio is current, and expected convergence exceeds all friction.

Best market state High liquidity, normalized spreads, limited idiosyncratic news, moderate intraday volatility, and a clear relative-value dislocation that is not a fundamental repricing.

2. Why the 10:30-14:30 Window Exists

The time window is a risk-control choice. The opening 30-60 minutes often includes overnight information processing, auction imbalance effects, stale quotes, delayed institutional participation, retail market orders, and news repricing. The closing window brings benchmark trading, rebalance flow, auction imbalances, and end-of-day risk management. A relative-value desk wants the cleaner middle: enough liquidity to trade both legs, but less one-way benchmark pressure.

Clock	Desk action	Reason
9:30-10:00	Observe and reject unstable pairs	Opening prints can create false residuals from stale quotes, spread widening, and delayed hedge updates.
10:00-10:30	Recalculate live beta, liquidity, z-score, and event filters	The market has begun to reveal whether the divergence is real, temporary, or news-driven.
10:30-11:30	Primary entry window	Enough price discovery has occurred; still enough time remains for intraday convergence.
11:30-13:30	Management and selective adds	Midday often rewards patient paired execution and cost control; avoid adding to broken relationships.
13:30-14:30	Harvest, reduce, or roll decision	Close converged spreads; convert remaining trades to explicit swing positions only if model supports it.
After 14:30	No new discretionary pairs unless separately tested	Closing flows can dominate residuals and distort convergence statistics.

Time-window rule

The same z-score has different meaning at 9:35, 12:00, and 15:45. Midday z-scores are more useful only if liquidity, beta, and event states are clean.

The point is not that 10:30-14:30 magically creates edge. The point is that it improves signal hygiene by avoiding the two most chaotic market microstructure periods of the U.S. cash session.

3. Research Basis and Professional Context

The academic lineage begins with distance-based pairs trading and expands into cointegration, stochastic spread models, factor residuals, PCA residuals, copulas, and high-frequency execution studies. The most important desk takeaway is that the strategy has evidence, but the edge is conditional and fragile.

Evidence anchor	Practical interpretation
Gatev, Goetzmann, and Rouwenhorst (2006)	Classic distance-based pairs trading found material average excess returns for self-financing pair portfolios, with profits above conservative transaction-cost estimates in their sample.
Avellaneda and Lee (2010)	Statistical arbitrage can be framed as trading mean-reverting idiosyncratic residuals after removing PCA or ETF sector exposures.
Do and Faff (2010)	After realistic costs, profitability persisted but at more modest levels. Pairs trading performed relatively well during turbulent periods.
Bowen, Hutchinson, and O Sullivan (2010)	High-frequency pairs profits are extremely sensitive to transaction costs and execution speed; delays can eliminate edge.
Krauss (2017)	The literature spans distance, cointegration, time-series, stochastic control, and machine-learning approaches; market frictions remain central.
Zhu (2024)	Recent replication found distance-based pairs still produced positive average excess return in the prior twenty years, while momentum/factor risk can materially impair returns.
Prop-desk analogy	Public descriptions from firms such as Jane Street, Citadel Securities, and Hudson River Trading emphasize quantitative models, correlated-asset risk, liquidity provision, and advanced execution infrastructure. Retail or independent traders must compensate with tighter selectivity and smaller capacity.

Professional relative-value desks do not rely on a single indicator. They build a full process: clean data, pair selection, model validation, risk neutrality, event filters, simultaneous execution, and constant post-trade review.

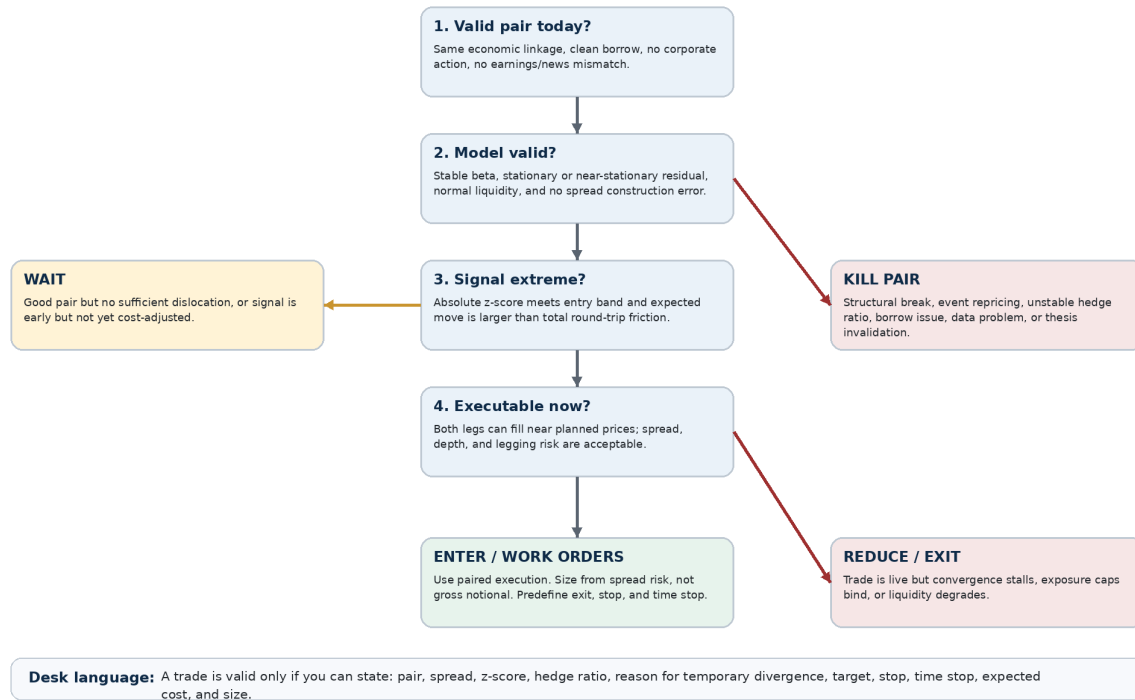
4. The Strategy in One Page

A production pairs/stat-arb trade should fit into one sentence:

"I am long the cheap leg and short the rich leg because the spread is temporarily dislocated versus a stable relationship; expected convergence exceeds round-trip friction; event risk is clean; beta/factor exposure is capped; and I am wrong if the spread breaks structure, beta becomes unstable, or time-to-convergence exceeds the stop."

Step	Question	Pass standard
1. Linkage	Why should these instruments move together?	Same industry, same factor exposure, ETF/constituent relationship, dual listing, or explicit economic linkage.
2. Model	How is the spread built?	Log price spread, regression residual, cointegration vector, factor residual, or dynamic beta residual.
3. Stationarity	Does the residual mean-revert?	Evidence from rolling stationarity tests, half-life, residual variance stability, and historical convergence.
4. Signal	Is the divergence large enough?	Absolute z-score exceeds threshold and expected convergence is large relative to costs.
5. Context	Is the divergence temporary?	No idiosyncratic news, earnings, borrow shock, merger rumor, litigation, index event, or sector repricing.
6. Execution	Can both legs be filled well?	Paired orders, limit logic, depth, low legging risk, and slippage below budget.
7. Risk	What kills the trade?	Predefined stop, time stop, exposure limit, and model-break rule.

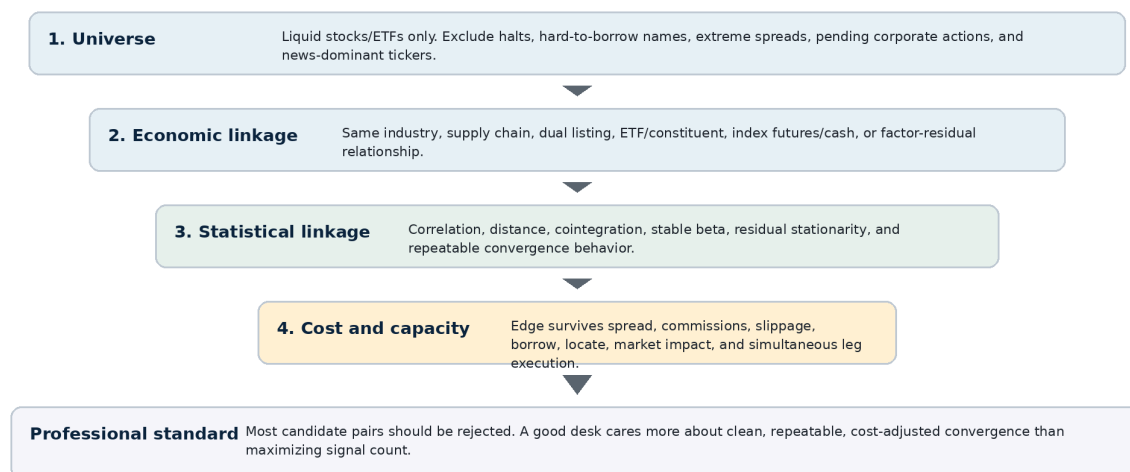
Live Decision Tree - Enter, Wait, Reduce, or Kill



5. Universe Construction

Most pair failures start before the signal. A weak universe creates false confidence because statistical similarity can be accidental. Build from tradable, economically related instruments first, then apply statistical filters.

Pair Selection Funnel - From Universe to Tradable Spread



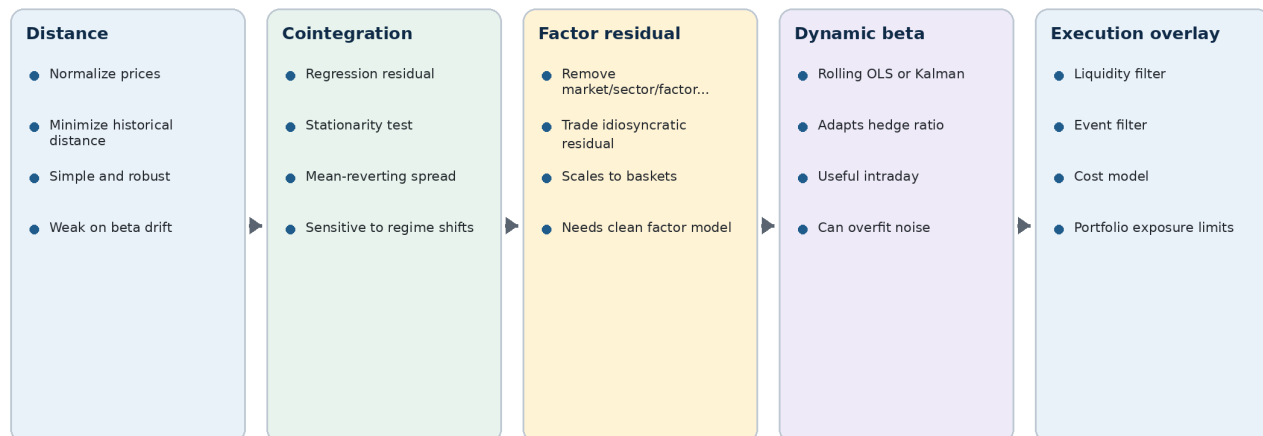
5.1 Hard universe filters

- Liquidity: minimum dollar volume, tight spread, stable depth, and sufficient midday participation to enter and exit both legs.
- Borrow: short leg must be borrowable with known fee and recall risk. Hard-to-borrow names require explicit risk premium.
- Corporate actions: exclude mergers, spin-offs, tender offers, index additions/removals, special dividends, and material legal/regulatory events unless explicitly modeled.
- Event calendar: avoid pairs where one leg has earnings, investor day, FDA decision, court ruling, or analyst event while the other does not.
- Price integrity: use clean midquotes or volume-weighted bars; avoid bid-ask bounce masquerading as mean reversion.

6. Pair Selection Methods

A robust desk uses more than one selection method. Distance finds close historical price paths. Cointegration tests whether a spread is stationary. Factor residuals remove common exposures. Dynamic beta adapts the hedge ratio intraday. The best pair candidates tend to pass multiple methods without requiring fragile parameter tuning.

Model Stack - Distance, Cointegration, Residual, and Dynamic Beta



Implementation rule Do not deploy a signal because the z-score is extreme. Deploy only after the model stack agrees that the spread is still a valid residual and the expected convergence can be captured after costs.

6.1 Method selection matrix

Method	Best use	Failure mode
Distance	Fast screening of large universes; simple normalized price-path similarity.	Can select pairs that look similar historically but lack economic linkage or stable hedge ratio.
Correlation	Sanity check for co-movement and factor linkage.	Correlation is not stationarity; two trending assets can be correlated without a tradable spread.
Cointegration	Pairs where a linear spread should mean-revert over the trading horizon.	Tests are sample-sensitive; stationarity can vanish after regime or business-model changes.
ETF/factor residual	Stock versus sector ETF, market beta, or multi-factor basket.	Poor factor model can leave hidden exposures or overfit historical noise.
Kalman beta	Intraday and dynamic relationships where hedge ratio drifts slowly.	May chase noise and create unstable sizing if not bounded.
Copula / nonlinear	Pairs with nonlinear dependence or tail relationship.	Complexity can hide overfitting and capacity limits.

7. Spread Construction

The spread is the tradeable object. A sloppy spread turns pairs trading into disguised directional betting. For intraday deployment, use midquotes where possible, time-align bars, remove bad prints, adjust for splits/dividends, and avoid using last trade prices when one leg is stale.

7.1 Core formulas

Log spread: $s_t = \log(P_{A,t}) - \beta_{A,t} * \log(P_{B,t})$
 Residual spread: $r_{A,t} = \alpha + \beta_{mkt} * MKT_t + \beta_{sec} * SEC_t + e_{A,t}$; trade $e_{A,t} - e_{B,t}$ or stock residual vs basket residual
 Z-score: $z_t = (s_t - \text{mean_window}(s)) / \text{std_window}(s)$
 Half-life: estimate $s_t = c + \phi * s_{(t-1)} + \text{error}$; $\text{half_life} = -\log(2) / \log(\phi)$, when $0 < \phi < 1$

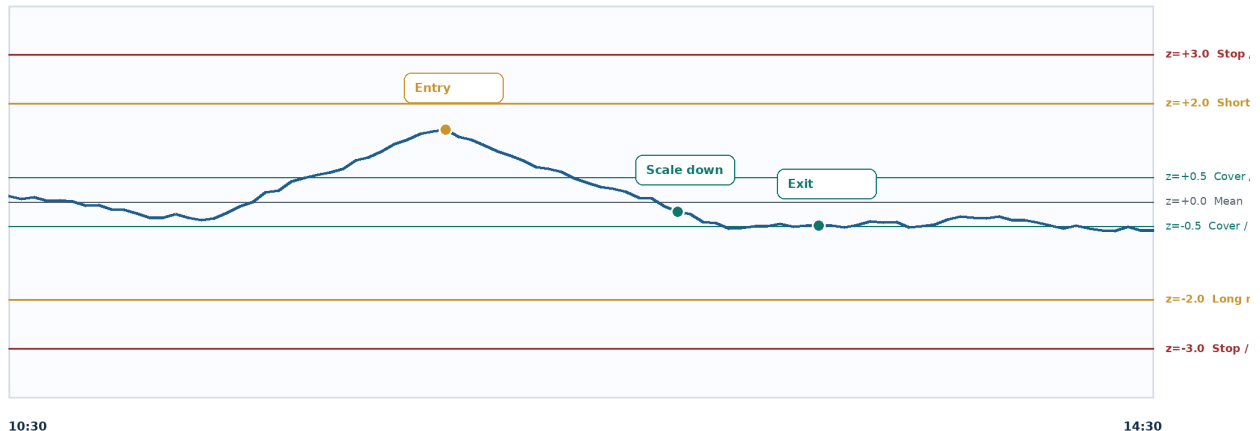
Spread choice	Professional use
Dollar spread	Easy to understand but unstable when prices differ or trend over time. Rarely sufficient alone.
Log spread	Common for equities because it scales price changes proportionally.
OLS residual	Good when one leg is the explanatory instrument and beta is stable over the sample.
Cointegration residual	Preferred when long-run equilibrium is statistically supported.
Factor residual	Preferred for basket/stat-arb where the alpha is idiosyncratic, not pair-specific.
Kalman residual	Useful when beta changes slowly and the system can bound beta drift.
No stale legs	If one leg trades frequently and the other is stale, the z-score can be fake. Use synchronized bars, midquotes, and freshness checks.

8. Z-Score Signal Design

Z-score converts a spread into standardized deviation. It is useful only if the spread distribution is stable. Advanced traders should treat z-score as a trigger candidate, not a trade instruction.

Spread Z-Score Anatomy - Entry, Exit, Stop, and Thesis Decay

Spread = $\log(A) - \beta \log(B)$. Z-score is only valid if the spread model is stable.



Expert interpretation: High absolute z-score is not sufficient. You trade only if the deviation is temporary, not caused by earnings, news, borrow stress, index inclusion, merger risk, sector repricing, or beta breakdown. Stops should react to structural invalidation, not just a bigger z-score.

8.1 Default z-score bands for research

State	Typical band	Action
Noise	$\text{abs}(z) < 1.0$	No trade. Let spread move or reassess model.
Watch	$1.0 \leq \text{abs}(z) < 1.75$	Prepare order ticket; require context confirmation.
Entry	$1.75 \leq \text{abs}(z) \leq 2.75$	Enter only if all filters pass and cost-adjusted expectancy is positive.
Extreme	$2.75 < \text{abs}(z) \leq 3.5$	Smaller size or wait for stabilization. Extremes can mean structural break.
Kill / review	$\text{abs}(z) > 3.5$	Do not add by default. Confirm whether relationship has broken.

Advanced rule

The more extreme the z-score, the less you should assume mean reversion. Extreme deviations often contain information.

9. Entry Playbooks

Entries must be paired, sized from spread risk, and filtered by live liquidity. The decision is not simply long/short; it is whether to work passively, cross immediately, split the order, or pass due to legging risk.

Setup	Entry trigger	Best conditions	Avoid when
Mean-reversion touch	Spread hits entry z-band and pauses without news.	Liquid pair, stable beta, good depth, low spread cost.	Deviation caused by one-leg news or sector repricing.
Failed continuation	Spread overshoots but cannot extend; price action shows exhaustion.	Extreme z-score, improving quote quality, fading order flow.	Momentum factor is strong and both legs are moving with sector.
VWAP-relative pair	One leg extended from VWAP relative to paired hedge; residual z-score agrees.	Both legs liquid intraday; sector ETF is stable.	One leg has event-driven volume.
ETF residual	Stock deviates from sector ETF/basket residual after open.	Broad market stable, clean sector benchmark, no idiosyncratic news.	Sector ETF itself is undergoing macro repricing.
Basket stat-arb	Stock residual extreme versus multi-factor basket.	Large universe, low idiosyncratic event risk, robust factor model.	Portfolio is crowded in one factor or sector.

9.1 Entry sequence

- Confirm model: hedge ratio is current, spread is stable, and residual z-score is not due to bad data.
- Check event tape: news, earnings, analyst notes, index events, borrow alerts, and halt risk are clean.
- Calculate net edge: expected convergence minus round-trip spread, commission, slippage, borrow, and market impact.
- Choose order style: midpoint/limit if passive edge is available; IOC/marketable only when missing the trade is worse than slippage.
- Execute both legs: avoid naked legging unless the desk explicitly models legging risk.

Order instruction

A professional ticket includes both legs, hedge ratio, target spread, stop spread, max slippage, time-in-force, and kill condition.

10. Execution Tactics for Both Legs

Execution is not an afterthought. In pairs trading, the edge may be smaller than the cost of a poor fill. The trader must manage slippage, spread crossing, queue priority, and legging risk.

Execution choice	When to use	Risk
Passive both legs	Spread is stable and queue position is likely.	Missed trade; adverse selection if you get filled on only the bad side.
Aggressive both legs	Signal is strong, capacity is limited, and convergence expected move is large.	Paying spread and impact can erase expectancy.
Passive cheap leg, aggressive rich leg	Cheap leg has favorable queue; rich leg is moving away.	Creates temporary directional exposure.
Spread/paired algorithm	System can route both legs simultaneously with slippage constraints.	Requires robust infrastructure and monitoring.
Staggered entries	Used only with pre-modeled legging risk and strict time limits.	Can become an unplanned directional trade.

10.1 Slippage budget

Expected gross convergence ≥ 2.5 to $4.0 \times$ expected round_trip_cost
 round_trip_cost = spread_A + spread_B + commissions + borrow + expected_market_impact + slippage_buffer

Capacity rule	If the trade only works at the screen midpoint and fails after paying realistic fills, it is not a professional trade.
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Expert traders should separate signal alpha from execution alpha. The signal says what should converge; execution determines whether the desk can actually capture it.

11. Trade Management

Once entered, the trade is managed at the spread level, not by emotional reaction to either single leg. The correct question is whether the residual thesis is improving, unchanged, or broken.

Management event	Action
Spread moves toward mean	Take partials at $z=0.5$ to 0.75 ; exit remainder near mean or when expected remaining move is smaller than exit cost.
Spread stalls near entry	Reduce if no progress by the expected half-life fraction; do not allow dead capital to become a hope trade.
Spread widens but model intact	Only add if pre-planned, within risk limits, and the wider price improves expectancy after costs.
Spread widens due to event	Exit or reduce immediately; statistical assumptions no longer apply.
One leg loses liquidity	Reduce or exit with priority on the more fragile leg.
Portfolio correlation spikes	Cut gross exposure even if individual pairs still look valid.
14:30 arrives	Close intraday-only trades or explicitly roll with overnight risk approval.

11.1 Exit hierarchy

- Primary target: spread returns to mean or to pre-defined exit band.
- Cost-adjusted target: exit when remaining expected convergence is less than exit friction plus risk charge.
- Time stop: exit if no convergence within expected half-life or by 14:30 for intraday mandate.
- Model stop: exit if stationarity, beta, liquidity, or event assumptions break.
- Portfolio stop: exit or reduce if sector/factor/drawdown limits bind.

Never average blindly

Adding to a widening spread without confirming the relationship is still valid is how market-neutral trades become large directional losses.

12. Stop Logic and Invalidation

Spread stops should combine price, statistics, time, and thesis. A pure price stop is too crude; a pure z-score stop can be too late. Professional desks use multiple kill conditions.

Stop type	Trigger	Reason
Spread stop	Spread breaches entry z-score by defined increment or breaks historical support/resistance.	Controls loss when convergence does not begin.
Model stop	Beta estimate jumps, residual volatility expands, stationarity fails, or data quality degrades.	The model no longer describes the relationship.
Event stop	News hits one leg, borrow changes, halt risk emerges, or analyst/filing catalyst appears.	The spread may be repricing fundamentals.
Time stop	No convergence within half-life threshold or by 14:30 for intraday trades.	Opportunity cost and model decay matter.
Liquidity stop	Spread widens, depth disappears, or expected exit cost exceeds remaining edge.	A good signal cannot compensate for untradeable liquidity.
Portfolio stop	Daily loss, factor exposure, sector concentration, or correlation cap is reached.	Protects the book from hidden common exposure.

12.1 Invalidation questions

- Would I enter this trade now if I had no position?
- Has one leg received new information that the other has not?
- Has the hedge ratio moved outside its historical confidence band?
- Is the z-score extreme because volatility compressed unrealistically before entry?
- Is the loss coming from the modeled residual or from unrecognized factor exposure?
- Can I exit both legs at a cost consistent with the original plan?

Expert discipline The stop is not where the pain is unbearable. The stop is where the thesis is no longer statistically or economically true.

13. Portfolio Construction and Neutrality

A book of individually neutral trades can still have large hidden exposure. Several pairs may all be long small-cap value, short mega-cap momentum, short hard-to-borrow names, or exposed to the same sector shock. Portfolio construction must be explicit.

Portfolio Risk Stack - Neutrality Is a Control System, Not a Label

Gross exposure	Limit total dollars at risk and total borrow usage. Gross can hide crowding and liquidity risk.
Net exposure	Keep dollar, beta, and sector net exposure within tested bands; do not assume equal dollars equals neutral risk.
Factor exposure	Monitor market, sector, size, value, momentum, volatility, and rates proxies where relevant.
Pair concentration	Cap exposure to a single pair, sector cluster, issuer, borrow pool, and shared factor driver.
Execution liquidity	Use capacity based on ADV, intraday volume curve, spread, depth, and expected participation rate.
Tail/event risk	Earnings, guidance, M&A, litigation, regulatory, index/rebalance, and borrow recalls override model signals.
Hard truth:	Pairs trades lose like concentrated directional trades when several pairs share the same hidden risk factor. Build exposure reports before increasing size.

13.1 Position sizing

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pair_risk = abs(current_spread - stop_spread) * spread_dollar_value
shares_A = target_gross / (P_A + abs(beta)*P_B)
shares_B = beta * shares_A * P_A / P_B    # adjust for dollar/beta convention
size = min(risk_budget / pair_risk, liquidity_capacity, borrow_capacity, portfolio_limit)

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Sizing rule	Size from spread loss to stop and liquidity capacity, not from confidence or how attractive the chart looks.
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14. Factor and Event Filters

The most dangerous pairs trades are not noisy; they are logically obsolete. A divergence caused by new information should not be faded as a statistical anomaly.

Filter	Pass requirement	Auto-reject condition
Earnings	No earnings today/after close for either leg unless both legs are modeled event trades.	One leg reports today or guidance changes materially.
News	No material single-name news explaining the spread.	M&A, lawsuit, product failure, regulatory action, fraud, restatement, FDA, activist, or downgrade with price impact.
Sector beta	Pair residual is not simply sector/factor repricing.	Spread move disappears after sector/factor adjustment or aligns with factor shock.
Borrow	Borrow cost and availability stable.	Hard-to-borrow shock, recall risk, locate uncertainty, or forced buy-in risk.
Index/rebalance	No special auction or benchmark flow driving one leg.	Index inclusion/removal, rebalance, unusually large close/open imbalance.
Liquidity	Current spread/depth is within tested bands.	Spread > 2x normal or depth disappears in one leg.

14.1 Event hierarchy

- Tier 1 - hard stop: earnings surprise, merger/tender offer, regulatory action, halt, fraud/restatement, borrow recall.
- Tier 2 - reduce or pause: analyst downgrade/upgrade, sector headline, macro shock, large ETF/index flow.
- Tier 3 - monitor: minor news, social attention, unusual volume without confirmed catalyst.

News beats statistics

When new information explains the divergence, the proper response is not to widen the stop. The proper response is to classify the pair as broken for that session.

15. Strategy Variants

Pairs trading is one member of a broader statistical-arbitrage family. Expert traders should choose the variant that matches the instrument relationship and available infrastructure.

Variant	Construction	When to implement
Classic pair	Long cheap leg, short rich leg using distance or cointegration spread.	Same-industry liquid stocks with clean economic link and stable beta.
ETF pair	Trade spread between related ETFs or ETF share classes.	Highly liquid, low-borrow, lower idiosyncratic risk; watch creation/redemption dynamics.
Stock vs sector ETF	Long/short stock against sector ETF hedge.	Single stock is temporarily mispriced versus sector; no stock-specific news.
Stock vs factor basket	Residual of a stock versus market/sector/factor peers.	Large universe, systematic pipeline, and factor risk reporting.
Index futures vs ETF	Relative-value spread between futures and ETF/cash proxy.	Requires futures pricing, carry, dividends, borrow, and low-cost execution.
Multi-leg stat-arb	Long/short basket of residuals ranked by model score.	Quant implementation with capacity, turnover, and factor constraints.
Expert selection If you cannot explain why the relationship should persist economically, do not rely on statistical similarity alone.		

For advanced discretionary traders, classic pairs and stock-vs-sector ETF residuals are the most practical. For systematic desks, basket residuals and multi-leg factor-neutral stat-arb generally scale better.

16. Backtesting Protocol

Backtests for pairs/stat-arb are easy to overstate. The strategy is highly sensitive to survivorship bias, stale prices, corporate actions, borrow, short constraints, costs, and signal timing. The test must imitate live decision-making.

Backtest requirement	Professional standard
Universe history	Survivorship-bias-free constituents with delisted stocks and real corporate actions.
Data timestamps	Use only information available at the decision time; no future membership, split, dividend, or earnings information.
Prices	Use bid/ask or midquotes with tradeability checks, not just last prices.
Costs	Commissions, spread, slippage, market impact, short borrow, locate cost, and failed borrow cases.
Execution delay	Include realistic delay between signal and fill; test immediate, one-bar, and adverse-fill assumptions.
Capacity	Limit participation by ADV and intraday volume curves; test performance decay with size.
Out-of-sample	Walk-forward pair selection, model calibration, threshold tuning, and validation.
Stress tests	Volatility spikes, correlation breakdowns, earnings seasons, crisis weeks, and sector rotations.

16.1 Minimum performance diagnostics

- Net expectancy after costs by time bucket: 10:30-11:30, 11:30-12:30, 12:30-13:30, 13:30-14:30.
- Win rate, average win/loss, profit factor, Sharpe, Sortino, drawdown, skew, and tail loss.
- MFE/MAE distribution by setup and by z-score band.
- Performance by sector, borrow type, market regime, volatility state, and news state.
- Slippage attribution by symbol, order type, spread, and liquidity bucket.
- Sensitivity to entry threshold, exit threshold, beta lookback, half-life, and costs.

Research rule

A backtest that ignores borrow, slippage, and simultaneous leg execution is an idea sketch, not an investable strategy.

17. Data Engineering Requirements

Data quality is edge preservation. Pairs signals are often small, and bad data can create more false signals than genuine opportunities. The live system should reject trades when data quality is uncertain.

Data field	Required treatment
Trades and quotes	Use NBBO or exchange-specific book; distinguish last trade from executable midquote.
Corporate actions	Split, dividend, special dividend, symbol change, merger, spin-off, and distribution adjustments.
Borrow and locate	Availability, fee, utilization, recall risk, and overnight holding permission.
Calendar	Earnings, macro events, dividends, index rebalances, ETF rebalance dates, and sector events.
Fundamentals/industry	Industry classification, revenue exposure, market cap, ADR/local relationship, and business-model similarity.
Factor returns	Market, sector, industry, size, value, momentum, volatility, and rates proxies.
Execution logs	Order timestamp, route, limit, fill, slippage, rejects, partial fills, and cancel/replace history.

17.1 Bad data kill-switches

- One leg has no fresh quote within the maximum freshness threshold.
- Bid/ask spread is outside normal percentile bands.
- Corporate-action adjustment changed historical spread shape unexpectedly.
- Data vendor reports crossed markets or missing quotes.
- The z-score is generated by one outlier print rather than sustained midquote movement.
- Live beta changes too much due to sparse data or quote anomalies.

System standard	The system should prefer false negatives to false positives. Missing a trade is acceptable; trading bad data is not.
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18. Live Desk Operating Schedule

A professional pairs desk separates preparation, observation, deployment, management, and review. The signal should not be discovered manually in the moment; it should be pre-ranked before the window opens.

Time	Action	Output
Pre-market	Refresh universe, corporate actions, borrow, earnings calendar, and prior-day pair rankings.	Approved watchlist and hard rejects.
9:30-10:00	Observe opening price discovery and remove event-driven pairs.	Clean live candidate set.
10:00-10:30	Recompute beta, spread z-score, liquidity, factor exposure, and cost-adjusted expectancy.	Ranked trade queue.
10:30-11:30	Enter top candidates that pass all gates.	Paired executions with full ticket metadata.
11:30-13:30	Manage, harvest, or add only if pre-authorized by model and risk.	Updated spread state and risk report.
13:30-14:30	Exit, reduce, or convert to explicit swing trade with overnight risk approval.	Closed or approved carry positions.
After close	Run post-trade analysis and update model diagnostics.	Review sheet and threshold changes.

18.1 Trade ticket template

Pair: A/B
 Direction: long cheap leg / short rich leg
 Spread model: $\log(A) - \beta \log(B)$, $\beta = \underline{\hspace{1cm}}$
 Entry z: $\underline{\hspace{1cm}}$ Target z: $\underline{\hspace{1cm}}$ Stop z: $\underline{\hspace{1cm}}$ Time stop: $\underline{\hspace{1cm}}$
 Expected cost: $\underline{\hspace{1cm}}$ bps Expected convergence: $\underline{\hspace{1cm}}$ bps
 Gross: $\underline{\hspace{1cm}}$ Net beta: $\underline{\hspace{1cm}}$ Sector exposure: $\underline{\hspace{1cm}}$
 Event status: clean / flagged / rejected
 Exit rule: target / model break / event / time / portfolio stop

19. Practical Example: Intraday Pair Trade

The following example is intentionally generic. It shows process, not a recommendation for any specific instrument.

Stage	Example decision
Pair	Two liquid same-industry large-cap stocks with historically stable beta and no earnings within five trading days.
Model	Log spread using rolling beta; spread passes stationarity and half-life screens on 60-day intraday history.
10:15 scan	Spread z-score = +2.1, meaning leg A is rich relative to leg B. Sector ETF is flat; no single-name news.
Cost model	Expected gross convergence = 38 bps of spread value; round-trip cost budget = 9 bps; trade passes 4:1 gross-to-cost rule.
Entry 10:42	Short rich leg A and long cheap leg B simultaneously with limit orders; partial passive fills allowed for 90 seconds only.
Management	At $z=1.0$ take one-third off; at $z=0.4$ take another third; exit remainder if z stalls or by 14:20.
Invalidation	Exit if z exceeds 3.0, beta estimate shifts beyond band, one leg receives news, or total pair loss reaches risk budget.
Post-trade	Record entry z , fill quality, MFE, MAE, exit reason, and whether the model or execution drove outcome.
Key lesson	The trade is not "stock A looks high and stock B looks low." The trade is "the residual spread is dislocated inside a validated relationship and the convergence can be captured after friction."

The example also shows why the 10:30-14:30 window matters. The trader uses the open to distinguish false dislocation from true relative-value opportunity, then enters only after live liquidity and event filters pass.

20. Common Failure Modes

Every mature pairs desk maintains a failure-mode library. The same mistakes repeat: trading correlation as if it were cointegration, underestimating costs, ignoring news, and turning a relative-value trade into a directional bet.

Failure mode	Symptom	Prevention
Correlation trap	Assets moved together historically but spread is not stationary.	Require spread stability, stationarity checks, and economic linkage.
Event repricing	One leg diverges for a real reason and keeps moving.	Real-time news/event filter and immediate model-break exit.
Hidden factor exposure	Many pairs lose together during factor rotation.	Run portfolio factor exposure and stress tests.
Cost illusion	Backtest profitable at mid, loses live.	Use bid/ask, slippage, borrow, and execution delay in research.
Legging loss	One side fills, the hedge side runs away.	Use paired execution, strict timeouts, and max leg imbalance.
Averaging down	Trader adds because z-score is bigger.	Add only when pre-planned and relationship remains valid.
Overfitting	Too many parameters work only in-sample.	Walk-forward validation and parameter stability tests.

20.1 Recovery protocol after a large loss

- Freeze new entries in the affected pair family or sector.
- Classify loss: model, event, execution, data, portfolio exposure, or discretionary override.
- Recompute factor and sector exposures at the time of loss.
- Review quote quality and slippage on both legs.
- Lower size or disable the setup until the failure source is corrected.
- Do not immediately widen thresholds to fit the loss.

Professional behavior

A model failure is information. Treat it as a research input, not as a personal challenge to win the money back.

21. Advanced Enhancements

Once the baseline process is profitable after costs, advanced enhancements can improve robustness. Add complexity only when it solves a measured problem.

Enhancement	Purpose	Guardrail
Kalman hedge ratio	Adapts beta to changing intraday relationships.	Bound beta and monitor overfitting/noise chasing.
Robust z-score	Uses median/MAD or EWMA volatility to reduce outlier sensitivity.	Validate against standard z-score and real fill outcomes.
Regime classifier	Switches thresholds by volatility, liquidity, trend, or macro regime.	Avoid tiny regime buckets and look-ahead bias.
Factor-neutral optimizer	Allocates across many residuals with exposure limits.	Constrain turnover and capacity.
Copula dependence	Models nonlinear dependence and tail behavior.	Ensure complexity improves out-of-sample net returns.
Meta-labeling	Predicts which z-score signals are worth taking.	Use strict cross-validation and leakage controls.
Execution learning	Improves fill tactics based on slippage history.	Do not optimize away risk controls to get better fills.

21.1 When to add complexity

- Only after the simple process is stable and fully costed.
- Only when post-trade diagnostics show a specific weakness to solve.
- Only if the enhancement improves out-of-sample net expectancy and drawdown.
- Only if live operators can understand the new failure modes.

Complexity rule	A more advanced model is not automatically a better trade. The best model is the simplest model that survives costs, regimes, and live execution.
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22. Pre-Trade, In-Trade, and Post-Trade Checklists

Expert execution is checklisted. The checklist reduces discretion where discretion is most dangerous: before entry, while losing, and after a profitable streak.

22.1 Pre-trade checklist

- Pair has economic linkage and is approved for today.
- Both legs are liquid, borrowable, and free of hard event conflicts.
- Hedge ratio is current and within acceptable bounds.
- Spread z-score is inside entry band and expected convergence exceeds round-trip cost by required multiple.
- Stationarity/half-life metrics remain valid for the intended holding period.
- Order plan includes both legs, max slippage, time-in-force, and cancel rules.
- Risk report confirms pair, sector, factor, gross, net, and daily drawdown limits.

22.2 In-trade checklist

- Spread is moving toward target or still valid within time stop.
- No new event explains the divergence.
- Liquidity remains tradable in both legs.
- Portfolio exposures remain within limits.
- No unplanned legging exposure persists beyond allowed time.
- Exit orders are prepared before target or stop is reached.

22.3 Post-trade checklist

- Was the trade model-valid at entry?
- Was the outcome caused by signal, execution, event, or portfolio exposure?
- Did actual slippage match the budget?
- Did the trader follow target, stop, and time-stop rules?
- Should this pair remain eligible tomorrow?

23. Default Research Parameters

The following defaults are starting points for research. They are not universal live settings. Convert all thresholds into symbol-specific and time-of-day-specific percentiles before deployment.

Parameter	Starting point for research
Universe	Top liquid stocks/ETFs by dollar volume; same-sector or economically linked only.
Formation window	20-60 trading days of intraday data plus longer daily sanity check.
Trading window	10:30-14:30 ET; no new intraday-only entries after 14:30.
Entry threshold	$\text{abs}(z)$ between 1.75 and 2.75 with context confirmation.
Exit threshold	z returns to 0.0-0.5 or expected remaining move falls below costs.
Stop threshold	z extends beyond 3.0-3.5 or model/event/liquidity stop triggers.
Half-life	Expected convergence should fit within the intended holding window; otherwise classify as swing trade.
Cost multiple	Expected gross convergence at least 2.5x-4.0x round-trip friction.
Risk per pair	Small fixed fraction of daily risk budget; reduce for correlated pair clusters.
Gross exposure cap	Defined by liquidity and drawdown tolerance, not by confidence.

23.1 Compact desk command language

"Pair A/B approved; model is log spread with beta 0.86; $z=+2.2$; short A long B; clean event tape; expected convergence 38 bps vs 9 bps cost; target $z=0.5$; stop $z=3.1$ or beta break; time stop 14:20; size 0.35R."

Pass rule

If the trader cannot state the spread model, hedge ratio, event status, expected cost, target, stop, and time stop before entry, the trade is not ready.

24. References and Evidence Base

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Final note

The literature supports conditional relative-value effects, not a universal edge. The production edge comes from clean relationships, rigorous cost modeling, fast rejection of broken pairs, and disciplined portfolio risk.

This playbook is educational and operational in nature. It is not personalized financial, legal, tax, or investment advice. Independent validation and risk review are required before live trading.